

## CISC 365 - Algorithms I

### Lecture 01: Introduction

- Your instruction team
- What is this course about? Why is it important?
- Learning expectations
- Course assessment
- Course management

School of Computing, Queen's University

1

### Your instructors

- Yuanzhu Chen (email: [yuanzhu.chen@queensu.ca](mailto:yuanzhu.chen@queensu.ca))
- Xu Wang (email: [xw41@queensu.ca](mailto:xw41@queensu.ca))
- Courses: CISC 102, CISC221, CISC335, CISC365
- Research interests:
  - Computer networking
  - Complex networks
  - Graph algorithms
  - Distributed AI
  - Data centre networking

2

2

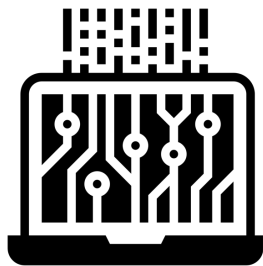
## What is computer science?

3

3

Computer science is the study of *algorithms*, including

- Their formal and mathematical properties
- Their hardware realizations
- Their linguistic realizations
- Their applications



Ethics

Applications

The software world

The virtual machine

The hardware

The algorithmic foundation

4

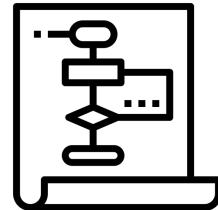
4

What is an *algorithm*?

5

5

## Algorithms



- Definition

- Informally, “an ordered sequence of instructions that is guaranteed to solve a specific problem”
- Formally, “a well-ordered collection of unambiguous and effective computable operations that, when executed, produces a result and halts in a finite amount of time”

6

6

## Algorithms in Computer Science

- Why are formal algorithms important?
  - If we can specify an algorithm to solve a problem, then we can automate its solution
  - We can build a machine, or write a program to carry out the steps
- Computer science is the science of ***algorithmic problem solving***

7

7

## Topics of This Course

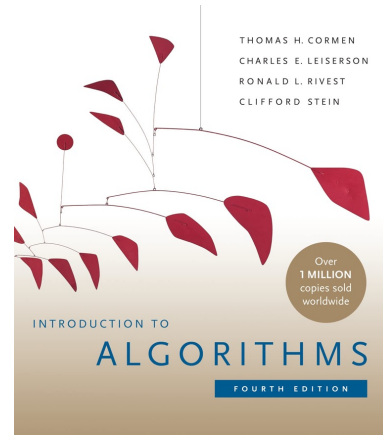
- Algorithm efficiency and problem complexity
- Divide-and-conquer
- Greedy strategies
- Dynamic programming
- Branch-and-bound

8

8

## Textbook, OnQ, and MS Teams

- *Introduction to Algorithms*, 4ed MIT Press (3ed is ok), by Cormen, Leiserson, Rivest, and Stein, 2022
- OnQ for all course material and announcement
- Course team for discussion and offline learning



9

9

## Lectures

- Based on select chapters of textbook
- Stirling Room B
  - Mondays 11:30-12:20
  - Tuesdays 13:30-14:20
  - Thursdays 12:30-13:20
- Slides
  - Available before lecture

10

## Class Schedule

| Week | Topics (Chapters)                     | Dates        | Evaluation |
|------|---------------------------------------|--------------|------------|
| 1    | 1a: Algorithm complexity (Ch 2)       |              |            |
| 2    | 1a: Algorithm complexity (Ch 3)       |              |            |
| 3    | 2: Divide-and-conquer (Ch 4)          | September 24 | Test 1     |
| 4    | 2: Divide-and-conquer (Ch 4)          |              |            |
| 5    | 3: Greedy strategy (Ch 15)            | October 8    | Test 2     |
| 6    | Fall reading break                    |              |            |
| 7    | 3: Greedy strategy (Ch 15)            |              |            |
| 8    | 4: Dynamic programming (Ch 14)        | October 29   | Test 3     |
| 9    | 4: Dynamic programming (Ch 14)        |              |            |
| 10   | 5: Branch-and-bound                   | November 12  | Test 4     |
| 11   | 5: Branch-and-bound                   |              |            |
| 12   | 1b: NP-completeness (Ch 34)           |              |            |
| 13   | 1b: NP-completeness (Ch 34)<br>review | December 1   | Test 5     |

11

## Learning expectations

- Know how to analyze the efficiency of an algorithm
- Know the major algorithmic paradigms
- Know how to design a correct and efficient algorithm

12

12

## Really, what you learn from this course

### 1. A head start on preparing for your technical interviews

- Pre-interview online assessment
- Online interview
- On-site interview
- Related coursework topics
- Write code and explain the design on the spot
- Related technical projects
- Show thought process of solving novel and unfamiliar problems



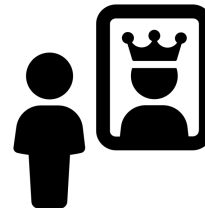
13

13

## Really, what you learn from this course

### 2. Training to solve hard and novel problems

- Active learning
- How to approach a new problem with learned techniques
- Self reflect
- Solving hard problems teaches us how to solve new hard problems
- Push your limit and learn from setbacks
- Build confidence and perseverance
- Stay optimistic
- Enjoy learning and growing



14

14

## Assessment



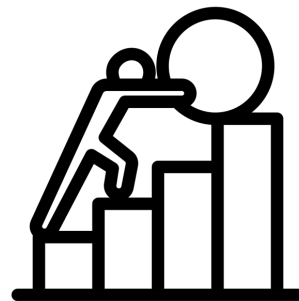
- Five tests (one test on each of the five topics)
  - 5 x 20%
- No mid-term and no final exam
- **No make-up tests**

15

15

## How to succeed in this course

- Come to the lectures
- Review each lecture and recreate the algorithms
- Self reflect after each chapter and test
- Get help
- Send us feedback



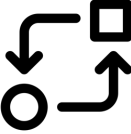
16

16

## My responsibility



Clarity and transparency



Adaptation



Help and guidance



Photo credit: <https://nathanielhoward.artstation.com/projects/kVwJl>

17

17

## Advice from last year students

- Attend lectures regularly
- Work on practice problems every week
- Utilize resources
- Start early and stay consistent
- Engage and collaborate
- Ask for help
- Mindset and strategy
- Prepare effectively for tests
- Take care of yourself
- General tips

18

18

## Teaching Team

### Instructors

Yuanzhu Chen

Xu Wang

### Teaching Assistants

Felix Xing

James Song

Mohamed Harmanani

### Our email

[cisc365@cs.queensu.ca](mailto:cisc365@cs.queensu.ca)

19

## Getting Help (in the following order)

1. **MS Teams discussion channels**
  - Public inquiries about course materials and tests through dedicated channels
  - The teaching team will monitor them and respond
2. **Email**
  - Private inquiries about course materials and assignments to [cisc365@cs.queensu.ca](mailto:cisc365@cs.queensu.ca)
  - Administrative and extenuating situations should be sent to Instructor directly
3. **Office hours**
  - With Instructor: Tuesdays 14:30-15:20pm
  - With TAs: TBA
4. **1:1 Appointments**
  - You can schedule 1:1 appointments when none of the above works for you

20

## Acknowledgement



Prof. Robin Dawes

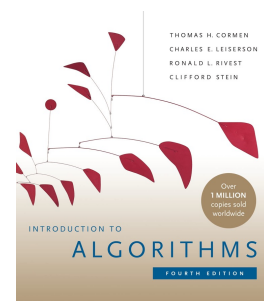


Prof. Ting Hu

21

## Reading

- Chapter 1: Role of Algorithms in Computing



22

22